

BHUVISION // SatQuery AI

Enterprise Production Deployment & Systems Engineering Manual

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1. Executive Summary & Production Topology

BHUVISION (SatQuery AI) is an autonomous multi-sensor satellite Earth intelligence platform designed to ingest and analyze multi-spectral optical imagery (Sentinel-2, Cartosat-3, Landsat-9) and Synthetic Aperture Radar (SAR Sentinel-1, RISAT-1B). The platform integrates a 9-agent autonomous reasoning graph, photorealistic 3D Google Earth terrain elevation, authenticated NASA GIBS and MapTiler feeds, Google Maps Platform hybrid layers, and a bank-grade multi-factor authentication suite.

Subsystem	Technology Component	Operational Function
Core Backend	FastAPI (Python 3.13) + Uvicorn + Pydantic v2	High-throughput async REST API, agent pipelines, telemetry
3D Google Earth Map	MapLibre GL 3.6.2 + MapTiler 3D + Three.js r128	60 FPS GPU-accelerated terrain elevation (DEM), 3D buildings, sky haze
Geospatial Feeds	Google Maps + MapTiler + NASA GIBS + AWS DEM	0.15m GSD Google Hybrid, 0.5m MapTiler, NRT 250m NASA, AWS Terrarium DEM
Agentic Reasoning	OpenRouter LLM Gateway (9 Specialized Models)	Gemini 2.5/2.0, Claude 3.7 Sonnet, Qwen 2.5 72B, DeepSeek V3, Llama 3.3
Authentication Suite	Google SSO OAuth 2.0 + Email/Pass + Phone SMS OTP	JWT Bearer tokens (24h TTL), bcrypt hashing, 6-digit cryptographic OTP
Mobile Distribution	Progressive Web App (PWA) + Manifest + Service Worker	Standalone mobile installability on Android, iOS, and Desktop

2. Authentication Suite Configuration

BHUVISION supports three enterprise authentication workflows: Google Single Sign-On (OAuth 2.0), Email/Password Registration, and Phone SMS OTP Verification.

2.1 Google OAuth 2.0 Setup

- Navigate to the Google Cloud Console (console.cloud.google.com) and create or select your project.
- Navigate to **APIs & Services** → **Credentials** and click **Create Credentials** → **OAuth Client ID**.
- Set Application Type to **Web application**.
- Under **Authorized JavaScript origins**, add your domains:

```
http://localhost:8000
http://127.0.0.1:8000
https://bhuvision.yourdomain.com
```

- Under **Authorized redirect URIs**, add:

```
https://bhuvision.yourdomain.com/api/auth/google/callback
```

- Copy your `Client ID` and `Client Secret` into your `.env` file:

```
GOOGLE_CLIENT_ID=your-google-client-id.apps.googleusercontent.com
GOOGLE_CLIENT_SECRET=GOCSPX-your-google-client-secret
```

2.2 Phone SMS OTP Integration

The platform includes endpoints for SMS dispatch (`/api/auth/otp/send`) and verification (`/api/auth/otp/verify`). By default, in local testing it uses secure simulated delivery. For production dispatch, configure your SMS Gateway provider (Twilio, AWS SNS, MSG91, or Fast2SMS):

```
SMS_GATEWAY_PROVIDER=twilio # options: twilio, aws_sns, msg91, fast2sms
TWILIO_ACCOUNT_SID=ACXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
TWILIO_AUTH_TOKEN=your_twilio_auth_token
TWILIO_PHONE_NUMBER=+1234567890
```

3. Geospatial & AI Gateways Configuration

3.1 Three Active Map API Providers

BHUVISION uniquely orchestrates three commercial and scientific map APIs concurrently:

- **Google Maps Platform:** Provides high-resolution Google Hybrid satellite imagery, vector maps, 3D tilt (67.5°), and geocoding. Enable *Maps JavaScript API* in GCP Console.
- **MapTiler Cloud:** Provides 3D Digital Elevation Models (Terrain-RGB) and satellite imagery. Configured via API Key in MapTiler Cloud Settings.
- **NASA Earthdata & GIBS:** Delivers near-real-time (NRT) 250m resolution MODIS Terra/Aqua and VIIRS daily reflectance layers via authenticated Bearer JWT tokens.
- **AWS Open Data Terrarium DEM:** Global open-access 3D elevation tiles used for zero-failure fallback.

3.2 OpenRouter 9-Agent Model Lineup

The multi-agent graph routes specialized subtasks to top-tier reasoning models via the OpenRouter gateway:

Agent Role	Configured LLM Model	Specialized Function
Lead Orchestrator	google/gemini-2.5-pro	High-level query decomposition and agent consensus
Query Planner	anthropic/claude-3.7-sonnet	Temporal bounding and sensor requirement planning
Sensor Router	meta-llama/llama-3.3-70b-instruct	Spectral band routing (Optical vs SAR vs NDVI)
SAR Radar Expert	deepseek/deepseek-r1	Dielectric roughness, double-bounce, flood backscatter
Change Detection	qwen/qwen-2.5-72b-instruct	Bi-temporal differencing and morphological analysis
Visual Grounding	google/gemini-2.0-flash-exp	Bounding polygon coordinates and MGRS projection
VQA Analyst	anthropic/claude-3.5-sonnet	Visual Question Answering on high-res tiles
Evidence Fusion	mistralai/mistral-large-2407	Multi-sensor synthesis and confidence calibration
Audit & Trace	openai/gpt-4o-mini	Immutable trace ledger and chain-of-custody

4. Step-by-Step Deployment Workflows

Option A: Docker & Docker Compose (Recommended for Production)

Deploy the entire stack with containerized FastAPI backend, Nginx reverse proxy, and automated SSL:

```
# 1. Clone repository on production server
git clone https://github.com/Ayushnot41/Satquery-Ai.git /opt/bhuvision
cd /opt/bhuvision

# 2. Configure production secrets
cp .env.example .env
nano .env # Add your domain, JWT secret, and API keys

# 3. Build and launch containerized services
docker compose up -d --build

# 4. Verify running health
docker compose ps
curl -I http://localhost:8000/api/health
```

Option B: Linux Cloud VM (Ubuntu 24.04 LTS / AWS EC2 / DigitalOcean)

Deploy as a systemd service behind an Nginx reverse proxy:

```
# 1. Install system prerequisites
sudo apt-get update && sudo apt-get install -y python3.13 python3.13-venv nginx certbot python3-certbot
-nginx git

# 2. Setup project virtual environment
cd /opt/bhuvision
python3.13 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r backend/requirements.txt

# 3. Configure Systemd Service (/etc/systemd/system/bhuvision.service)
sudo tee /etc/systemd/system/bhuvision.service << 'EOF'
[Unit]
Description=BHUVISION SatQuery AI FastAPI Application
After=network.target

[Service]
User=www-data
WorkingDirectory=/opt/bhuvision/backend
ExecStart=/opt/bhuvision/venv/bin/uvicorn app.main:app --host 127.0.0.1 --port 8000 --workers 4
Restart=always

[Install]
WantedBy=multi-user.target
EOF

# 4. Start and enable service
sudo systemctl daemon-reload
sudo systemctl start bhuvision
sudo systemctl enable bhuvision
```

5. Nginx Reverse Proxy & SSL Configuration

Create an Nginx configuration block with HTTP/2, WebSocket proxying, and caching:

```
server {
    listen 80;
    server_name bhuvision.yourdomain.com;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl http2;
    server_name bhuvision.yourdomain.com;

    ssl_certificate /etc/letsencrypt/live/bhuvision.yourdomain.com/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/bhuvision.yourdomain.com/privkey.pem;

    # Gzip Compression
    gzip on;
    gzip_types text/plain text/css application/json application/javascript text/xml;

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

```
# Issue free Let's Encrypt SSL Certificate
sudo certbot --nginx -d bhuvision.yourdomain.com
```

6. Mobile Web App (PWA) Deployment

BHUVISION is fully engineered as an installable Progressive Web App:

- **Web App Manifest:** Hosted at /manifest.json with display: standalone, theme_color: #0B1120, and complete ISRO icon sets.
- **Android Chrome Installation:** Open the URL on mobile Chrome → tap the three dots → tap **"Add to Home Screen"** or **"Install App"**. Runs full-screen without URL bars.
- **iOS Safari Installation:** Open on mobile Safari → tap the **Share** button → tap **"Add to Home Screen"**. Launches as a native-feel iOS web app.
- **Google Play Store Packaging:** Use Microsoft PWABuilder (pwabuilder.com) or Bubblewrap CLI to generate an Android signed .aab package directly from your live URL.

7. Automated Testing & Verification Suite

Before releasing to production, execute the automated verification suite:

```
# Execute complete pytest suite (43 passed tests)
pytest backend/tests/ -v --tb=short

# Verify SVG asset integrity
pytest backend/tests/test_svg_assets_validity.py -v

# Health check probe
curl -s http://127.0.0.1:8000/api/health
```

Production Security Checklist:

1. Set `ENVIRONMENT=production` and `DEBUG=false` in `.env`.
2. Change the default `JWT_SECRET_KEY` to a high-entropy 256-bit random string.
3. Ensure `CORS ORIGINS` in `main.py` only allow authorized production hostnames.
4. Restrict Google Cloud and MapTiler API keys with HTTP referer domain restrictions.